

most mammals grow two sets of teeth in their lifetime

the large ribcage provides space for big lungs and a heart to power the active body

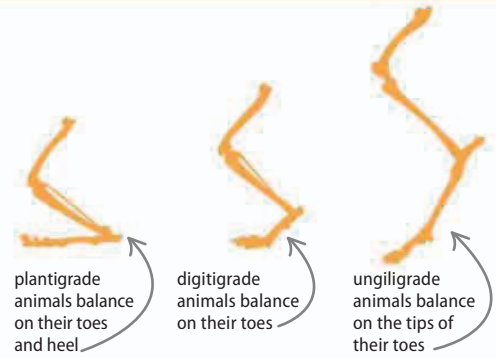
▷ Mammal body

Most mammals are quadrupeds (they walk on four legs), although humans, kangaroos, and bears are able to walk on two feet. Mammals' legs are directly beneath the body, not to the side as in reptiles. This allows them to walk long distances and run at high speeds.

the tail bones are an extension of the spine

feet have claws (or toenails or hooves in other mammals) made from keratin

dog



△ Stances

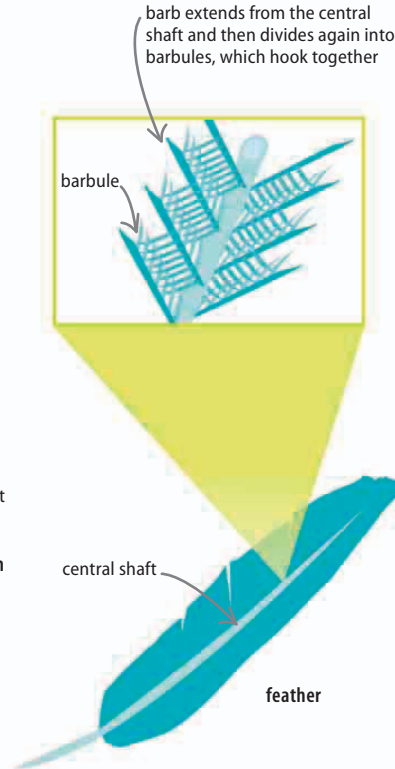
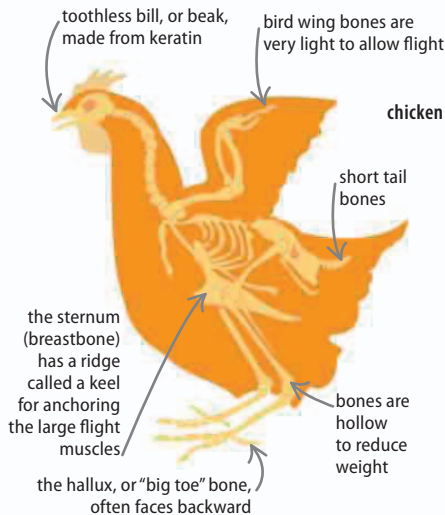
Mammals stand in three ways. Animals that walk long distances, such as humans and bears, are plantigrade. Digitigrade feet are used by agile animals that run and jump, such as dogs. Unguligrade animals, such as horses, are suited to high-speed running.

Birds

The first birds evolved from forest-living dinosaurs about 150 mya. With about 10,000 species known, birds are the dominant flying vertebrates. Their wings are formed from long feathers attached to the bones of the forelimb. Feathers are stiff but lightweight, making them ideal for forming a rigid flight surface.

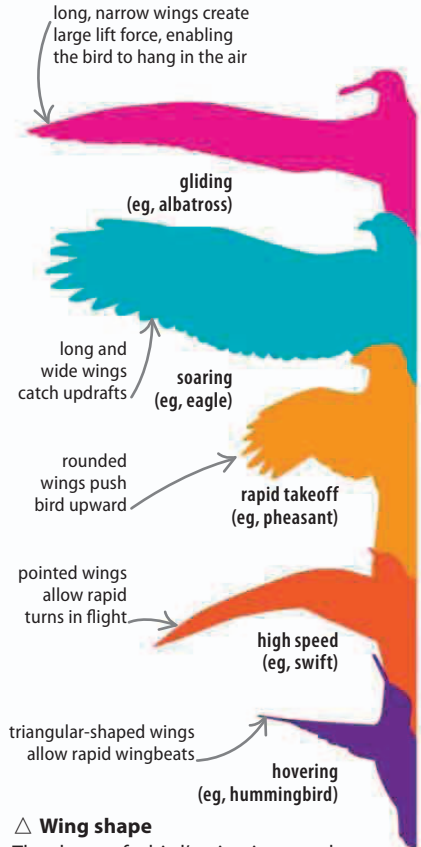
▽ Bird skeleton

Birds evolved from bipedal (upright-walking) dinosaurs. The wing is formed from the forelimb with thickened finger bones extending from the end to increase the length.



△ Feather anatomy

A feather is made of a branching network of hooked keratin filaments. Birds must frequently preen, applying oils that keep the filaments clean and hooked together into a flat surface.



△ Wing shape

The shape of a bird's wing is a good indicator of how it flies. Scavenging birds need long, curved wings to glide, while ground birds need short wings to take off and get away from predators quickly.